THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 14: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following cannot be the probability of an event? 1

(a) 0.99 **(b)** $\frac{3}{5}$ **(c)** 0.4 **(d)** $\frac{7}{6}$

2. A die is thrown once. The probability of getting a number greater than 4 is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{6}$

3. A die is thrown once. The probability of getting a number between 2 and 5 is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{1}{2}$

4. A coin is tossed two times. The probability of getting exactly one tail is: 1

(a) $\frac{2}{5}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{4}$

5. A bag contains 7 red, 3 blue balls. One ball is drawn at random. The probability that it is blue is: 1

(a) $\frac{3}{10}$ **(b)** $\frac{2}{5}$ **(c)** $\frac{1}{5}$ **(d)** $\frac{7}{10}$

6. 20 cards numbered 1 to 20 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a multiple of 4 is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{3}{10}$

7. The probability of an event E is $\frac{5}{12}$. The probability of the event 'not E' is: 1

(a) $\frac{17}{12}$ **(b)** $\frac{7}{12}$ **(c)** $\frac{1}{12}$ **(d)** $\frac{5}{12}$

8. Which of the following cannot be the probability of an event? 1

(a) $\frac{2}{9}$ **(b)** -0.3 **(c)** 1 **(d)** 0

9. A die is thrown once. The probability of getting a prime number is: 1

(a) $\frac{3}{7}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{2}{3}$

10. A die is thrown once. The probability of getting a number less than 7 is: 1

(a) 1 **(b)** $\frac{5}{6}$ **(c)** 0 **(d)** $\frac{6}{7}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A bag contains 4 white, 6 black balls. One ball is drawn at random. Find the probability that it is black. 2

12. A die is thrown once. Find the probability of getting a number that is both prime and even. 2

13. A coin is tossed two times. Find the probability of getting exactly one tail. 2

14. A bag contains 8 yellow, 2 green balls. One ball is drawn at random. Find the probability that it is green. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. In a lottery there are 6 prizes and 14 blanks. A ticket is drawn at random. Find the probability of getting a prize. 3

16. A lot of 120 ball pens contains 18 defective pens. One pen is drawn at random from the lot. Find the probability that the pen is good (not defective). 3

17. A class has 45 students, of whom 18 are girls. A student is selected at random to be the monitor. Find the probability that a boy is selected. 3

OR

In a lottery there are 9 prizes and 21 blanks. A ticket is drawn at random. Find the probability of getting a prize. 3

18. A lot of 60 ball pens contains 9 defective pens. One pen is drawn at random from the lot. Find the probability that the pen is good (not defective). 3

OR

A class has 25 students, of whom 10 are girls. A student is selected at random to be the monitor. Find the probability that a boy is selected. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A bag contains 9 red, 6 blue, 5 green balls. One ball is drawn at random. Find the probability that the ball is: 5
(i) blue,
(ii) neither red nor blue,
(iii) red or green (in this order).

OR

A box contains 25 cards numbered 1 to 25. One card is drawn at random. Find the probability that the number is: 5
(i) a perfect square,
(ii) a number divisible by 5,
(iii) a number greater than 25 (in this order).

20. A bag contains 4 red, 8 blue, 8 green balls. One ball is drawn at random. Find the probability that the ball is: 5
(i) not red,
(ii) blue,
(iii) red, blue or green (in this order).

OR

A box contains 30 cards numbered 1 to 30. One card is drawn at random. Find the probability that the number is: 5
(i) divisible by both 2 and 3,
(ii) a number divisible by 5,
(iii) a prime number (in this order).

END OF QUESTION PAPER Total: 40 Marks